FEM for the Navier–Stokes Equations

Personal Study Guide

Based on prerequisite lectures (Num. Math. II) and Chapter 2 slides

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Notation: Conventions used in this document

This guide follows the notation of the **Num. Math. II** lectures (Breiten & May). The Chapter 2 slides (Münch) use slightly different symbols. A translation table is provided in Section 10.

Concept	Num. Math. II	Chapter 2 slides
Basis / shape functions	ψ_i	φ_j
Discrete subspace	V_n	V_h
Discrete solution	u_n	u_h
Unknown coefficients	d_i	U_j
Galerkin / stiffness matrix	A_n	K
Right-hand-side vector	$\mathbf{f}$	F

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Chapter 1: The Big Picture

The entire FEM pipeline answers one question: **how do we turn a PDE (a continuous, infinite-dimensional problem) into a linear system $A_n\,\mathbf{d} = \mathbf{f}$ that a computer can solve?**

Key Idea: The FEM Pipeline

$$\underset{\text{strong form}}{\boxed{-\Delta u = f}} \xrightarrow[\text{integrate}]{\text{multiply by } v} \underset{\text{weak form}}{\boxed{a(u,v) = \ell(v)}}$$

$$\xrightarrow[\text{dim. reduction}]{\text{Galerkin}} \underset{\text{discrete form}}{\boxed{a(u_n,v_n) = \ell(v_n)}} \xrightarrow[u_n = \sum d_i\psi_i]{\text{basis expansion}} \underset{\text{linear system}}{\boxed{A_n\,\mathbf{d} = \mathbf{f}}}$$

Every concept below is a piece of this chain.

Chapter 2: The Model Problem: Poisson Equation

2.1 Strong form

Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with boundary $\partial\Omega = \Gamma_D \cup \Gamma_N$, $\Gamma_D \cap \Gamma_N = \emptyset$.

Definition: Poisson equation — strong form

Find $u : \bar{\Omega} \to \mathbb{R}$ such that

$$-\Delta u = f \quad \text{in } \Omega, \tag{1}$$

$$u = g \quad \text{on } \Gamma_D \quad \text{(Dirichlet b.c.)}, \tag{2}$$

$$\frac{\partial u}{\partial n} = h \quad \text{on } \Gamma_N \quad \text{(Neumann b.c.)}. \tag{3}$$

Here $\Delta u = \sum_{i=1}^{d} \frac{\partial^2 u}{\partial x_i^2}$ is the Laplacian and $\frac{\partial u}{\partial n} = \nabla u \cdot \mathbf{n}$ is the outward normal derivative.

Intuition: What does the Poisson equation model?

Think of u as temperature. The source term f describes where heat is generated or absorbed. The Laplacian Δu measures how much u at a point differs from its average over nearby points. The equation says: at steady state, the curvature of u is balanced by the source f.

Dirichlet b.c. ($u = g$): we prescribe the *value* on part of the boundary (e.g. fixed temperature).

Neumann b.c. ($\partial u/\partial n = h$): we prescribe the *flux* on the rest (e.g. insulated wall $\Rightarrow h = 0$).

Chapter 3: From Strong to Weak Form

Source: Num. Math. II, Lecture 12.

3.1 Why we need a weak form

The strong form (1) requires $u \in C^2(\Omega) \cap C(\bar{\Omega})$, i.e. u must be twice continuously differentiable. This is too restrictive:

- Materials with discontinuous properties (e.g. a dielectric constant that jumps) produce solutions with "kinks" (Lecture 15, num_math).
- The weak form only needs *first* derivatives, admitting rougher but physically meaningful solutions.

3.2 Derivation (4 steps)

Definition: Function spaces

The **Sobolev space** $H^1(\Omega)$ contains all functions that are square-integrable together with their first (weak) derivatives:

$$H^1(\Omega) = \{\, v : \Omega \to \mathbb{R} \mid v \in L^2(\Omega),\ \nabla v \in [L^2(\Omega)]^d \}.$$

Important subspaces used in the lectures:

$$\begin{aligned} C_0^2(\bar{\Omega}) &= \{\, v \in C^2(\bar{\Omega}) \mid v(0) = 0 = v(1) \} && \text{(Lecture 12: first test space)}, \\ V_0 &= \{\, v \in H^1(\Omega) \mid v = 0 \text{ on } \Gamma_D \} && \text{(general Dirichlet test space)}, \\ V &= \{\, v \in H^1(\Omega) \mid v = 0 \text{ on } \Gamma_D \} && \text{(trial space, when } g = 0\text{; otherwise lifted)}. \end{aligned}$$

Starting from $-\Delta u = f$ in Ω:

Step 1: Multiply by a test function and integrate. Let $v \in V_0$ (so $v = 0$ on Γ_D). Multiply both sides by v and integrate over Ω:

$$\int_\Omega (-\Delta u)\, v \,\mathrm{d}x = \int_\Omega f\, v \,\mathrm{d}x.$$

Step 2: Integration by parts (Green's first identity).

Key Idea: The central trick

Integration by parts shifts one derivative from u onto v. Recall the multivariate product rule (Lecture 12, p. 4): $\operatorname{div}(v\,\nabla u) = \nabla u \cdot \nabla v + v\,\Delta u$. Integrating and applying Gauss's theorem:

$$\int_\Omega (-\Delta u)\, v \,\mathrm{d}x \;=\; \underbrace{\int_\Omega \nabla u \cdot \nabla v \,\mathrm{d}x}_{\text{only first derivatives!}} \;-\; \underbrace{\int_{\partial\Omega} v \cdot \frac{\partial u}{\partial n} \,\mathrm{d}A}_{\text{boundary integral}}.$$

This is why the weak form needs only first derivatives of both u and v.

Step 3: Apply boundary conditions. Split the boundary integral over Γ_D and Γ_N:

- On Γ_D: $v = 0$ by choice of test space $\Rightarrow$ integral vanishes.
- On Γ_N: $\partial u/\partial n = h$ (Neumann data) $\Rightarrow$ integral becomes $\displaystyle\int_{\Gamma_N} h\, v \,\mathrm{d}s$.

(Lecture 12, p. 4 shows this step-by-step for the 2D case with $\Gamma = \Gamma_1 \cup \Gamma_2$.)

Step 4: Write the weak form.

Definition: Weak / variational formulation (VP)

Find $u \in V$ such that

$$\boxed{a(u,v) = \ell(v) \qquad \forall\, v \in V_0} \tag{4}$$

where the **bilinear form** and **linear functional** are:

$$a(u,v) = \int_\Omega \nabla u \cdot \nabla v \,\mathrm{d}x, \tag{5}$$

$$\ell(v) = \int_\Omega f\, v \,\mathrm{d}x + \int_{\Gamma_N} h\, v \,\mathrm{d}s. \tag{6}$$

Intuition: Why does weak = strong?

Lecture 12 proves this using the **Fundamental Lemma of Calculus of Variations**: if $\int_\Omega g(x)\, h(x)\,\mathrm{d}x = 0$ for all smooth h with compact support, then $g \equiv 0$.
Applied here: define $g(x) = -u''(x) - f(x)$. The weak form says $\int g\, v\,\mathrm{d}x = 0$ for all $v \in C_0^\infty$. By the lemma, $g \equiv 0$, i.e. $-u'' = f$ pointwise—recovering the strong form. So the two formulations are **equivalent** whenever u is smooth enough.

Note: Neumann b.c. are "natural"

Neumann conditions appear *automatically* in the weak form via the boundary integral—you do *not* enforce them explicitly. Dirichlet conditions, by contrast, are *essential*: they are enforced by restricting the function space (Lecture 17, num_math).

Chapter 4: The Galerkin Method

Source: Num. Math. II, Lecture 14.

4.1 Three-step recipe

Step 1 (known): Find the variational / weak form. Find $u \in V$ such that $a(u, v) = \ell(v)$ for all $v \in V$. The space V is infinite-dimensional—we cannot work with it directly on a computer.

Step 2 (new): Galerkin dimension reduction. For each $n \in \mathbb{N}$, let $V_n \subset V$ be a subspace with $\dim(V_n) = n$. Pick a basis $\{\psi_1, \dots, \psi_n\}$ of V_n.

Definition: Galerkin equation

Find $u_n \in V_n$ such that

$$a(u_n, v_n) = \ell(v_n) \qquad \forall\, v_n \in V_{0,n} \subset V_0.$$

Step 3 (new): Derive a linear system of equations. Since V_n is finite-dimensional, every element can be written as

$$u_n(x) = \sum_{i=1}^{n} d_i\, \psi_i(x),$$

where $d_1, \dots, d_n \in \mathbb{R}$ are the unknown coefficients.

Key Idea: Testing with basis functions ψ_j is enough

Instead of testing with arbitrary $v_n \in V_n$, it suffices to test with each basis function ψ_j, $j = 1, \dots, n$.

Why? Any $v_n = \sum_j \beta_j\, \psi_j$. By linearity of a and ℓ:

$$a(u_n, v_n) = \ell(v_n) \implies \sum_j \beta_j \left[a(u_n, \psi_j) - \ell(\psi_j)\right] = 0.$$

Since the β_j are arbitrary, each bracket must vanish individually.

Substituting $u_n = \sum_i d_i\, \psi_i$ and testing with ψ_j:

$$a\Big(\sum_{i=1}^{n} d_i\, \psi_i,\ \psi_j\Big) = \ell(\psi_j) \qquad \xRightarrow{\text{bilinearity}} \qquad \sum_{i=1}^{n} d_i\, a(\psi_i, \psi_j) = \ell(\psi_j).$$

This is a linear system of n equations for n unknowns:

$$\boxed{A_n\, \mathbf{d} = \mathbf{f}} \tag{7}$$

with

$$(A_n)_{ji} = a(\psi_i,\, \psi_j) = \int_\Omega \nabla\psi_i \cdot \nabla\psi_j \,\mathrm{d}x, \tag{8}$$

$$f_j = \ell(\psi_j) = \int_\Omega f\, \psi_j \,\mathrm{d}x + \int_{\Gamma_N} h\, \psi_j \,\mathrm{d}s. \tag{9}$$

$A_n \in \mathbb{R}^{n\times n}$ is the **Galerkin matrix** (also called **stiffness matrix**), $\mathbf{f} \in \mathbb{R}^n$ is the **right-hand-side (load) vector**, and $\mathbf{d} = (d_1, \dots, d_n)^T$ are the unknown **degrees of freedom** (DOFs).

4.2 Galerkin orthogonality and best approximation

Source: Num. Math. II, Lecture 14, p. 2.

Since the true solution u of (VP) satisfies $a(u, v) = \ell(v)$ for all $v \in V$, in particular for all $v_n \in V_n \subset V$:

$$a(u, v_n) = \ell(v_n) = a(u_n, v_n) \qquad \forall\, v_n \in V_n.$$

Subtracting:

Definition: Galerkin orthogonality

$$\boxed{a(u - u_n,\ v_n) = 0 \qquad \forall\, v_n \in V_n} \tag{10}$$

The error $u - u_n$ is "a-orthogonal" to the entire discrete subspace V_n.

Intuition: Best approximation

If a is symmetric and positive definite, Galerkin orthogonality means u_n is the **closest** element in V_n to the true solution u, measured in the **energy norm** $\|v\|_a = \sqrt{a(v, v)}$:

$$\|u - u_n\|_a = \inf_{v_n \in V_n} \|u - v_n\|_a .$$

The FEM does not give just *some* approximation—it gives the **best possible** one from the chosen subspace.

4.3 Properties of the bilinear form

Source: Num. Math. II, Lecture 14, p. 2.

Definition: Bilinear form properties

Let V be an $\mathbb{R}$-vector space and $a : V \times V \to \mathbb{R}$ a bilinear form. We call a:

- **symmetric**, if $a(u, v) = a(v, u)$ for all $u, v \in V$;
- **positive definite**, if $a(u, u) \geq 0$ for all u and $a(u, u) = 0$ if and only if $u = 0$;
- **negative definite**, if $-a$ is positive definite.

If a is positive definite, the variational problem has a **unique** solution.

Note: Positive definiteness depends on the space!

Lecture 14 (p. 3) gives a crucial example: $a(u, v) = \int_\Omega \nabla u \cdot \nabla v \, \mathrm{d}x$.

1. On $V = \{v \in H^1(\Omega) \mid \partial v/\partial n = 0 \text{ on } \partial\Omega\}$: a is **not** positive definite (constant functions have $\nabla u = 0$ but $u \neq 0$).
2. On $V_0 = \{v \in H^1(\Omega) \mid v = 0 \text{ on } \partial\Omega\}$: a **is** positive definite (Dirichlet b.c. kill the constants).

The choice of boundary conditions determines whether the problem is well-posed!

Chapter 5: Finite Elements: The Choice of V_n

Source: Num. Math. II, Lectures 15–16.

The Galerkin method works for any subspace V_n. The key question is: how to choose V_n (and hence $\psi_1, \dots, \psi_n$) so that (a) the approximation is accurate, (b) A_n is sparse and easy to assemble, and (c) V_n behaves well as $n \to \infty$?

The answer from Lecture 15: **hat functions** (piecewise linear polynomials on a mesh).

5.1 Admissible decomposition (mesh)

Definition: Admissible decomposition

Decompose $\bar{\Omega} = \bigcup_{i=1}^{n_e} \bar{\Omega}_i$ into elements. The decomposition is **admissible** if for each pair $i \neq j$ exactly one of the following holds:

1. $\bar{\Omega}_i \cap \bar{\Omega}_j = \emptyset$ (disjoint),
2. $\bar{\Omega}_i \cap \bar{\Omega}_j$ is a complete shared face/edge/vertex.

- In 1D ($d = 1$): non-overlapping intervals $\Omega_\ell = (x_{\ell-1}, x_\ell)$, $\ell = 1, \dots, n_e = N + 1$.
- In 2D ($d = 2$): triangles or convex quadrilaterals with compatible edges (Lecture 15, p. 3 illustrates admissible vs. non-admissible decompositions).

5.2 Shape functions (basis functions ψ_i)

5.2.1 1D: hat functions (Lectures 15–16)

For a 1D mesh $a = x_0 < x_1 < \cdots < x_N < x_{N+1} = b$, the **linear Lagrange basis** ("hat functions" / "pyramid functions") are:

$$\psi_i(x) = \begin{cases} \dfrac{x - x_{i-1}}{x_i - x_{i-1}} & x \in [x_{i-1},\, x_i], \\ \dfrac{x_{i+1} - x}{x_{i+1} - x_i} & x \in [x_i,\, x_{i+1}], \\ 0 & \text{otherwise}, \end{cases} \qquad i = 1, \dots, N.$$

Properties (Lecture 16, p. 1):

- $\psi_i(x_j) = \delta_{ij}$ (equals 1 at node i, zero at all other nodes).
- ψ_i is piecewise linear and continuous; $\psi_i \in H^1(\Omega)$ but $\psi_i \notin C^1(\bar{\Omega})$.
- Each ψ_i has **local support**: nonzero only on the two neighboring elements.
- $\Rightarrow$ $(A_n)_{ij} = a(\psi_i, \psi_j) = 0$ whenever $|i - j| > 1$ $\Rightarrow$ A_n is **tridiagonal**.

5.2.2 Explicit stiffness matrix (1D, equidistant mesh, homogeneous Dirichlet)

Source: Num. Math. II, Lecture 16, p. 1.

For $\Omega = (0, 1)$, uniform mesh width $h = 1/(N + 1)$, and $u(0) = u(1) = 0$:

$$(A_n)_{ij} = \int_0^1 \psi_i'(x)\, \psi_j'(x)\, \mathrm{d}x = \frac{1}{h} \begin{cases} 2 & \text{if } i = j, \\ -1 & \text{if } |i - j| = 1, \\ 0 & \text{otherwise.} \end{cases}$$

$$A_n = \frac{1}{h} \begin{pmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & \ddots \\ & & -1 & 2 \end{pmatrix} \in \mathbb{R}^{N \times N}.$$

The right-hand side is typically computed by numerical quadrature: $f_j = \ell(\psi_j) = \int_0^1 f(x)\, \psi_j(x)\, \mathrm{d}x \approx \sum_q w_q\, f(x_q)\, \psi_j(x_q)$.

Note: Same as finite differences!

The FEM matrix $A_n = \frac{1}{h}\,\mathrm{tridiag}(-1, 2, -1)$ is identical to the finite-difference matrix from Lecture 01. For linear elements in 1D the two methods coincide. This is *not* true in 2D or for higher-order elements.

5.2.3 2D element catalog (Lecture 16, pp. 2–3)

On each element $\bar{\Omega}_i$, the local basis function is a polynomial. Common FEM choices:

Element type	**Cell shape**	**Nodes**	**DOFs/elem.**
Linear (P_1)	triangle	3	3
Quadratic (P_2)	triangle	6	6
Cubic (P_3)	triangle	10	10
Bilinear (Q_1)	quadrilateral	4	4
Biquadratic (Q_2)	quadrilateral	9	9

- Triangle with degree p: $\binom{p+2}{2}$ nodes (polynomial $\psi_i|_{\bar{\Omega}_i} = a_0 + a_1 x_1 + a_2 x_2 + \cdots$).
- Quadrilateral with degree p: $(p+1)^2$ nodes (tensor-product polynomial).
- The **Lagrange polynomial** formula $\ell_i(x) = \prod_{k \neq i} \frac{x - c_k}{c_i - c_k}$ guarantees $\ell_i(c_j) = \delta_{ij}$ (Lecture 16, p. 2).

Note: Dimension of $V_n \neq$ DOFs/element $\times$ number of elements

The dimension of the resulting subspace V_n depends on the *global* node count and the boundary conditions, not simply on the per-element DOF count (Lecture 16, p. 3).

Chapter 6: Assembly: Building A_n and f

6.1 Element-wise assembly

We never compute $(A_n)_{ij}$ by integrating over all of Ω at once. Since shape functions have local support, each integral reduces to a sum over element contributions:

Key Idea: Assembly via element matrices

$$A_n = \sum_c R_c^T A_c R_c, \qquad \mathbf{f} = \sum_c R_c^T \mathbf{f}_c,$$

where:

- c loops over all cells (elements) in the mesh,
- A_c is the **element stiffness matrix** (small, e.g. 3×3 for a P_1 triangle),
- $\mathbf{f}_c$ is the **element load vector**,
- R_c is the **restriction matrix** mapping local DOF indices to global ones.

6.2 Integration by numerical quadrature

The integrals $(A_c)_{ij} = \int_{\Omega_c} \nabla\psi_j \cdot \nabla\psi_i \, \mathrm{d}x$ are evaluated numerically on a reference element $\hat{\Omega}$:

$$\int_{\Omega_c} g(x) \, \mathrm{d}x \approx \sum_{q=1}^{n_q} w_q \, g(\mathbf{x}_q) \, |\det J|,$$

where $\{\mathbf{x}_q, w_q\}$ are the quadrature points and weights, and J is the Jacobian of the mapping $\hat{\Omega} \to \Omega_c$.

6.3 Boundary conditions in practice

Neumann: Already incorporated in $\ell(v)$—nothing extra to do.

Dirichlet ($u = g$ on Γ_D): For each DOF i on Γ_D:

1. Set row i of A_n to zero except $(A_n)_{ii} = 1$.
2. Modify the other rows: $f_j \leftarrow f_j - (A_n)_{ji} \, g_i$ for each $j \neq i$.
3. Set $f_i = g_i$.

This enforces $d_i = g_i$ while preserving the solution at interior nodes.

Chapter 7: Error Estimates

7.1 Interpolation error

Definition: Polynomial interpolation error

Given $f \in C^{n+1}[a,b]$ and its interpolating polynomial p_n of degree n at $n+1$ distinct points:

$$e_n(x) = f(x) - p_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}\,\omega_{n+1}(x), \quad \omega_{n+1}(x) = \prod_{k=0}^{n}(x - x_k).$$

For FEM shape functions of polynomial degree p on elements of size h:

$$\|\nabla e\|_{L^2} \le \frac{C(p)}{p!}\,h^p.$$

7.2 A priori error estimate

Combining the interpolation error bound with Galerkin orthogonality (10):

Key Idea: A priori convergence rate

For polynomial degree p and mesh size h:

$$\boxed{\|\nabla(u - u_n)\|_{L^2} \le C\,h^p \left\|\nabla^{p+1} u\right\|_{L^2}} \tag{11}$$

- $p = 1$ (linear elements): error $\sim \mathcal{O}(h)$. Halve the mesh $\Rightarrow$ halve the error.
- $p = 2$ (quadratic elements): error $\sim \mathcal{O}(h^2)$. Halve the mesh $\Rightarrow$ quarter the error.

Note: Regularity matters

The estimate requires $u \in H^{p+1}$ (the solution must be smooth enough). On domains with re-entrant corners (e.g. L-shaped), the solution may have singularities that limit the actual convergence rate regardless of p—motivating adaptive mesh refinement.

7.3 A posteriori error estimate

Given a *computed* solution u_n, estimate the error *without* knowing u:

$$\|e_n\|_A \le C\sqrt{\sum_T \Big(h_T^2\,\|R_T\|_T^2 + h_T\,\|J_E\|_{\partial T}^2\Big)}, \tag{12}$$

where:

- $R_K = f + \Delta u_n$ is the **cell residual** (how well u_n satisfies the PDE inside each element),
- $J_E = -\frac{1}{2}[\![\partial_n u_n]\!]$ is the **edge residual** (jump in normal derivative across element boundaries).

This drives **adaptive mesh refinement (AMR)**:

$$\text{solve} \;\to\; \text{mark cells with large error indicators} \;\to\; \text{refine} \;\to\; \text{repeat}.$$

Chapter 8: Theory: Existence and Uniqueness

Source: Num. Math. II, Lecture 17.

Definition: Continuity and coercivity

Let $\Omega \subset \mathbb{R}^d$ be a bounded polygonal domain and set $V = H_0^1(\Omega)$ with norm $\|u\|_V = \|u\|_{H^1(\Omega)}$. Assume $a : V \times V \to \mathbb{R}$ and $\ell : V \to \mathbb{R}$ satisfy:

a) **Continuity of a:** $\exists\, \alpha > 0$ such that $|a(v_1, v_2)| \le \alpha \, \|v_1\|_V \, \|v_2\|_V \;\; \forall\, v_1, v_2 \in V$.
b) **Coercivity (V-ellipticity) of a:** $\exists\, \beta > 0$ such that $a(v, v) \ge \beta \, \|v\|_V^2 \;\; \forall\, v \in V$.
c) **Continuity of ℓ:** $\exists\, g > 0$ such that $|\ell(v)| \le g \, \|v\|_V \;\; \forall\, v \in V$.

Theorem 8.1 (Lax–Milgram)**.** *Under assumptions (a)–(c), the variational problem (4) has a* ***unique*** *solution $u \in V$ with the stability estimate*

$$\|u\|_V \le \frac{g}{\beta}.$$

Intuition: Lax–Milgram ≈ invertibility in disguise

Compare with linear algebra (Lecture 17, p. 2):

	Linear algebra	**Variational problem**
Problem	$Ax = b$	$a(u, v) = \ell(v)$
Bilinear	$a(x, y) = y^T A x$	$a(u, v)$
Continuity	$\lvert y^T A x\rvert \le \|A\| \, \|x\| \, \|y\|$	$\lvert a(u, v)\rvert \le \alpha \, \|u\| \, \|v\|$
Coercivity	$x^T A x \ge \beta \, \|x\|^2$ (pos. def.)	$a(v, v) \ge \beta \, \|v\|^2$
Solution	$x = A^{-1} b$, $\|x\| \le \|A^{-1}\| \, \|b\|$	$\|u\|_V \le g/\beta$

Coercivity is the infinite-dimensional version of positive definiteness.

Note: a is *not required to be symmetric* for Lax–Milgram (Lecture 17, p. 2).

8.1 Variational ⇔ Minimization (when a is symmetric)

Source: Num. Math. II, Lecture 15, p. 1.

Theorem 8.2 (Equivalence of VP and MP)**.** *Let a be symmetric and positive definite. Define $\mathcal{J} : V \to \mathbb{R}$ by*

$$\mathcal{J}(v) = \tfrac{1}{2}\, a(v, v) - \ell(v).$$

Then the variational problem (VP) and the minimization problem

$$\text{(MP)} \quad \text{Find } u_{\min} \in V \text{ such that } \mathcal{J}(u_{\min}) \le \mathcal{J}(v) \;\; \forall\, v \in V$$

have the same (unique) solution.

Proof sketch (Lecture 15): (VP)⇒(MP): Define $w = u_{\min} + \epsilon\, v$ and compute $\tilde{\mathcal{J}}(\epsilon) = \mathcal{J}(u_{\min} + \epsilon v)$. Minimum at $\epsilon = 0$ implies $\tilde{\mathcal{J}}'(0) = 0$, which yields $a(u_{\min}, v) = \ell(v)$. The reverse direction follows similarly.

Chapter 9: Extension to Higher Dimensions

The variational framework is **dimension-independent**. Moving from 1D to 2D/3D requires changing:

1. **Mesh:** triangles / quadrilaterals (2D), tetrahedra / hexahedra (3D).
2. **Shape functions ψ_i:** e.g. linear on triangles (3 nodes), bilinear on quads (4 nodes).
3. **Quadrature rules:** for numerical integration on the reference element.
4. **Reference-to-physical mapping:** Jacobian J for coordinate transformation.

Key consequence: in 2D with unstructured meshes, A_n is still **sparse** (each ψ_i overlaps only with neighbors), but the sparsity pattern is less regular than the clean banded structure in 1D.

Chapter 10: Notation Reference

Term	Symbol	Meaning
Trial space	V	Space where we seek u (infinite-dim.)
Test space	V_0	Functions v we multiply by; $v = 0$ on Γ_D
Discrete subspace	V_n	Finite-dim. subspace of V; $\dim(V_n) = n$
Basis functions	$\psi_1, \ldots, \psi_n$	Piecewise polynomials; basis of V_n
Unknown coefficients	$d_1, \ldots, d_n$	DOFs: $u_n = \sum d_i \, \psi_i$
Discrete solution	u_n	Approximation in V_n
Bilinear form	$a(\cdot, \cdot)$	"Stiffness" part: $\int \nabla u \cdot \nabla v \, \mathrm{d}x$ for Poisson
Linear form	$\ell(\cdot)$	"Load" part: $\int f \, v \, \mathrm{d}x + \int_{\Gamma_N} h \, v \, \mathrm{d}s$
Galerkin matrix	A_n	$(A_n)_{ji} = a(\psi_i, \, \psi_j)$; stiffness matrix
Mass matrix	M_n	$(M_n)_{ij} = \int \psi_i \, \psi_j \, \mathrm{d}x$; appears in time-dep. problems
RHS vector	$\mathbf{f}$	$f_j = \ell(\psi_j)$; load vector
Galerkin orthogonality		$a(u - u_n, \, v_n) = 0 \;\; \forall \, v_n \in V_n$
Lax–Milgram		Continuity + coercivity $\Rightarrow$ unique solution
A priori estimate		Error bound in terms of h, p, and regularity of u
A posteriori estimate		Error bound computable from u_n alone
AMR		Adaptive mesh refinement: solve $\to$ estimate $\to$ refine $\to$ repeat

Chapter 11: Study Roadmap

Recommended reading order for the prerequisite lectures:

1. **Lecture 12** (pp. 1–4): Weak form derivation—understand every line, especially the integration-by-parts step on p. 4.
2. **Lecture 14** (pp. 1–2): Galerkin 3-step recipe ($V \to V_n \to A_n\mathbf{d} = \mathbf{f}$). Galerkin orthogonality and uniqueness.
3. **Lecture 14** (p. 3): Positive definiteness depends on the space—work through the two examples.
4. **Lecture 15** (last 2 pages): Admissible decompositions and hat functions—the bridge from Galerkin to FEM.
5. **Lecture 16** (p. 1): Explicit 1D stiffness matrix—compute A_n by hand for $N = 3$.
6. **Lecture 16** (pp. 2–3): 2D element catalog: P_1, P_2, Q_1, Q_2, Lagrange polynomials.
7. **Lecture 17** (p. 1): Neumann BCs in weak form.
8. **Lecture 17** (pp. 2–3): Lax–Milgram theorem and the analogy with linear algebra.

Self-test: Starting from $-\Delta u = f$, can you explain why "multiply by v and integrate by parts" ultimately produces the matrix equation $A_n\,\mathbf{d} = \mathbf{f}$? If yes, you are ready for the lecture.